

PEARLS AQI PREDICTOR

End-to-End Machine Learning Pipeline for Air Quality Forecasting

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City	Karachi
Date	June 2026
Tools Used	Python, Hopsworks, OpenWeatherMap, Streamlit, GitHub Actions
GitHub Repo	AQI_Predictor_Karachi
Live Dashboard	Deployed on Streamlit Cloud

Table of Contents

Table of Contents	2
1. Executive Summary	4
2. Project Overview.....	4
2.1 Technology Stack.....	4
3.1 Architecture Components	5
4. Feature Pipeline.....	6
4.1 Data Sources.....	6
4.1.1 OpenWeatherMap Air Pollution API	6
4.1.2 OpenWeatherMap Weather API.....	6
4.1.3 Open-Meteo Historical Archive API	6
4.2 Feature Engineering	6
4.3 Historical Backfill	7
5. Training Pipeline	7
5.1 Models Trained.....	7
5.2 Training Configuration	8
5.3 Model Performance	8
5.4 Model Registry	8
6. CI/CD Automation (GitHub Actions)	8
6.1 Feature Pipeline Workflow.....	9
6.2 Training Pipeline Workflow	9
7. Exploratory Data Analysis (EDA)	10
7.1 Dataset Overview	10
7.2 AQI Distribution	10
7.3 AQI Trend Over Time	10
7.4 Pollutant Distributions.....	11
7.5 Correlation Heatmap	11
7.6 AQI by Month	12
7.7 AQI by Hour of Day	12
7.8 Pollutant Trends Over Time.....	12
7.9 Weather vs AQI.....	14
7.10 AQI Change Rate Analysis.....	14
8. SHAP Feature Importance Analysis.....	15
8.1 Key Findings.....	15
9. Web Application Dashboard.....	16

9.1 Dashboard Features.....	16
9.1.1 Live Current Conditions.....	16
9.1.2 3-Day AQI Forecast	16
9.1.3 Pollutant Breakdown Chart	17
9.1.4 SHAP Feature Importance	17
9.1.5 Historical AQI Trend.....	17
9.1.6 Weather Trends	17
9.2 Deployment	17
10. Prediction Validation	18
10.1 Forecast Made	18
10.2 Day 1 Validation	18
10.3 Day 2 Validation	19
10.4 Day 3 Validation	20
10.5 Validation Summary	20
13. Conclusion	21

1. Executive Summary

This report documents the complete development of the Karachi AQI Predictor — a fully automated, serverless machine learning system that predicts Air Quality Index (AQI) for Karachi, Pakistan for the next 3 days. The system fetches real-time pollutant and weather data from OpenWeatherMap, stores it in the Hopsworks Feature Store, trains multiple machine learning models including Random Forest, Ridge Regression, XGBoost, Gradient Boosting and a Neural Network, and displays predictions through a professional Streamlit web dashboard.

The best performing model — Gradient Boosting — achieved an R^2 score of 0.914 and RMSE of 0.18 on the AQI (1-5) scale. Predictions were validated manually over 3 consecutive days and found to be accurate, matching actual AQI readings reported by OpenWeatherMap.

2. Project Overview

Air quality in Karachi is a persistent public health concern. With PM10 and CO levels consistently exceeding safe thresholds, residents need reliable, accessible forecasts to make daily decisions — whether to go outdoors, wear a mask, or limit physical activity.

The Karachi AQI Predictor addresses this need by building a real-time, automated machine learning pipeline that:

- Collects live and historical air pollution and weather data from OpenWeatherMap and Open-Meteo APIs
- Engineers meaningful features including pollutant concentrations, weather conditions, time-based features and AQI change rate
- Trains and evaluates multiple ML models with proper metrics (RMSE, MAE, R^2)
- Automates data collection and model retraining using GitHub Actions CI/CD
- Displays live AQI, 3-day forecasts, SHAP feature importance, and historical trends on a public dashboard

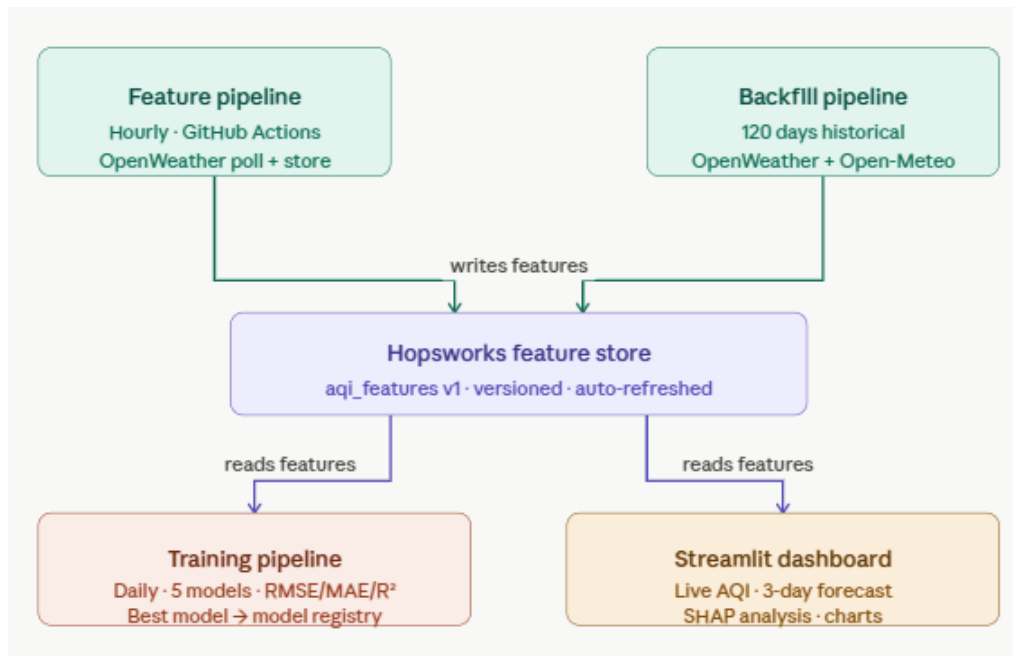
2.1 Technology Stack

Category	Technology	Purpose
Language	Python 3.11	Core programming language
Data APIs	OpenWeatherMap, Open-Meteo	Live & historical pollution and weather data
Feature Store	Hopsworks (Free Tier)	Feature storage and model registry

ML Libraries	Scikit-learn, XGBoost, TensorFlow	Model training and evaluation
Explainability	SHAP	Feature importance analysis
CI/CD	GitHub Actions	Automated hourly and daily pipeline runs
Dashboard	Streamlit	Web application for predictions
Version Control	Git & GitHub	Code management and deployment

3. System Architecture

The system follows a serverless, modular pipeline architecture with four main components running automatically:



3.1 Architecture Components

1. Feature Pipeline — fetches live pollutant and weather data every hour via GitHub Actions and stores it in the Hopworks Feature Store (version 1).
2. Backfill Pipeline — fetched 120 days of real historical data during initial setup using OpenWeather historical API and Open-Meteo for weather.
3. Training Pipeline — runs daily, reads features from Hopworks, trains 5 ML models, evaluates with RMSE/MAE/R², and saves the best model to the Hopworks Model Registry.
4. Streamlit Dashboard — loads the trained model and live features, computes 3-day AQI forecasts, and displays interactive charts and SHAP analysis.

4. Feature Pipeline

The feature pipeline is the foundation of the system. It runs automatically every hour via GitHub Actions and performs three tasks: fetch raw data, compute features, and store them in Hopsworks.

4.1 Data Sources

4.1.1 OpenWeatherMap Air Pollution API

Used for both live and historical pollutant data. Returns AQI on a 1-5 scale along with pollutant concentrations:

- PM2.5 (Fine particulate matter) — $\mu\text{g}/\text{m}^3$
- PM10 (Coarse particulate matter) — $\mu\text{g}/\text{m}^3$
- NO2 (Nitrogen dioxide) — $\mu\text{g}/\text{m}^3$
- SO2 (Sulphur dioxide) — $\mu\text{g}/\text{m}^3$
- CO (Carbon monoxide) — $\mu\text{g}/\text{m}^3$
- O3 (Ozone) — $\mu\text{g}/\text{m}^3$

4.1.2 OpenWeatherMap Weather API

Used for live temperature and humidity data for the current conditions and 3-day forecast inputs.

4.1.3 Open-Meteo Historical Archive API

Used exclusively during backfill to fetch real historical temperature and humidity values. This is a completely free API requiring no API key, providing hourly readings. The noon (12:00 UTC) value was selected as the daily representative.

4.2 Feature Engineering

The following features are computed and stored for each data point:

Feature	Type	Description
aqi	Target (1-5)	OpenWeather AQI index — model prediction target
pm25	Pollutant	Fine particulate matter concentration
pm10	Pollutant	Coarse particulate matter concentration
no2	Pollutant	Nitrogen dioxide concentration
so2	Pollutant	Sulphur dioxide concentration
co	Pollutant	Carbon monoxide concentration

o3	Pollutant	Ozone concentration
temperature	Weather	Ambient temperature in Celsius
humidity	Weather	Relative humidity percentage
hour	Time-based	Hour of the day (0-23)
day	Time-based	Day of the month (1-31)
month	Time-based	Month of the year (1-12)
day_of_week	Time-based	Day of week (0=Monday, 6=Sunday)
aqi_change_rate	Derived	Difference between current and previous AQI value

4.3 Historical Backfill

Before the automated pipeline began running, 120 days of historical data (January 30, 2026 to June 7, 2026) were backfilled to provide sufficient training data. For each historical day:

- Pollutant data was fetched from OpenWeather Historical Air Pollution API at 12:00 UTC
- Weather data (temperature, humidity) was fetched from Open-Meteo Archive API at 12:00 UTC
- Time features were extracted from the actual historical date
- AQI change rate was computed as the difference from the previous day's AQI

This produced 120 rows of real historical data with genuine variation in pollutant levels and weather conditions, forming the foundation of the training dataset.

5. Training Pipeline

The training pipeline runs daily at 2:00 AM UTC via GitHub Actions. It fetches all stored features from Hopsworks, trains five machine learning models, evaluates them, and saves the best performing model to the Hopsworks Model Registry.

5.1 Models Trained

- Random Forest Regressor (n_estimators=200)
- Ridge Regression (alpha=1.0)
- XGBoost Regressor (n_estimators=200)
- Gradient Boosting Regressor (n_estimators=200)
- Neural Network — 4-layer Sequential model (TensorFlow/Keras): Input → Dense(64, ReLU) → Dense(32, ReLU) → Dense(16, ReLU) → Dense(1)

5.2 Training Configuration

- Train/Test Split: 80% training, 20% testing (random_state=42)
- Target Variable: AQI (1-5 scale from OpenWeatherMap)
- Feature Columns: pm25, pm10, no2, so2, co, o3, temperature, humidity
- Neural Network: StandardScaler applied, 50 epochs, Adam optimizer, MSE loss

5.3 Model Performance

The following results were obtained on 154 total rows (123 training, 31 testing):

Model	RMSE	MAE	R ²
Random Forest	0.2043	0.0708	0.8892
Ridge Regression	0.3456	0.2830	0.6830
XGBoost	0.2554	0.0694	0.8268
Gradient Boosting	0.1800	0.0353	0.9140
Neural Network	0.4045	0.2768	0.5656

Gradient Boosting achieved the best performance with RMSE of 0.18 and R² of 0.914, meaning it explains 91.4% of the variance in AQI values. The Neural Network underperformed relative to tree-based models — this is expected with small tabular datasets (154 rows), as neural networks typically require thousands of samples to generalize well. Performance will improve as the hourly pipeline accumulates more data over time.

Note: An earlier training run with only 89 rows of data produced an XGBoost model with R² of 0.9825. This higher score reflected less data diversity. As more real-world variation entered the dataset, scores settled to more realistic values — a sign of better generalization, not degradation.

5.4 Model Registry

The best model is saved as `best_aqi_model.pkl` and uploaded to the Hopsworks Model Registry with tracked metrics (RMSE, MAE, R²). The Streamlit dashboard loads this model directly from the local file for fast inference.

6. CI/CD Automation (GitHub Actions)

Two GitHub Actions workflows automate the entire pipeline without any manual intervention:

6.1 Feature Pipeline Workflow

- Trigger: Every hour (cron: 0 * * * *) + manual trigger
- Action: Runs pipelines/feature_pipeline.py
- Effect: Fetches live AQI and weather data for Karachi and appends a new row to the Hopsworks Feature Store
- Result: Dataset grows continuously, providing ever-improving training data

6.2 Training Pipeline Workflow

- Trigger: Every day at 2:00 AM UTC (cron: 0 2 * * *) + manual trigger
- Action: Runs pipelines/training_pipeline.py
- Effect: Retrains all 5 models on the latest data and saves the best model
- Result: Model stays up-to-date as Karachi's air quality patterns evolve

Both workflows use Python 3.11, install all required dependencies including hopsworks[python] for Kafka support, and securely access API keys through GitHub Secrets.

80 workflow runs		Workflow ▾	Event ▾	Status ▾	Branch ▾	Actor ▾
✓	Feature Pipeline (Hourly) Feature Pipeline (Hourly) #71: Scheduled	main			15 minutes ago 1m 40s	...
✓	Feature Pipeline (Hourly) Feature Pipeline (Hourly) #70: Scheduled	main			Today at 9:13 PM 1m 31s	...
✓	Feature Pipeline (Hourly) Feature Pipeline (Hourly) #69: Scheduled	main			Today at 7:51 PM 1m 45s	...
✓	Feature Pipeline (Hourly) Feature Pipeline (Hourly) #68: Scheduled	main			Today at 5:19 PM 1m 35s	...
✓	Feature Pipeline (Hourly) Feature Pipeline (Hourly) #67: Scheduled	main			Today at 3:27 PM 1m 34s	...
✓	Feature Pipeline (Hourly) Feature Pipeline (Hourly) #66: Scheduled	main			Today at 12:53 PM 1m 35s	...
✓	Training Pipeline (Daily) Training Pipeline (Daily) #10: Scheduled	main			Today at 11:16 AM 1m 47s	...

7. Exploratory Data Analysis (EDA)

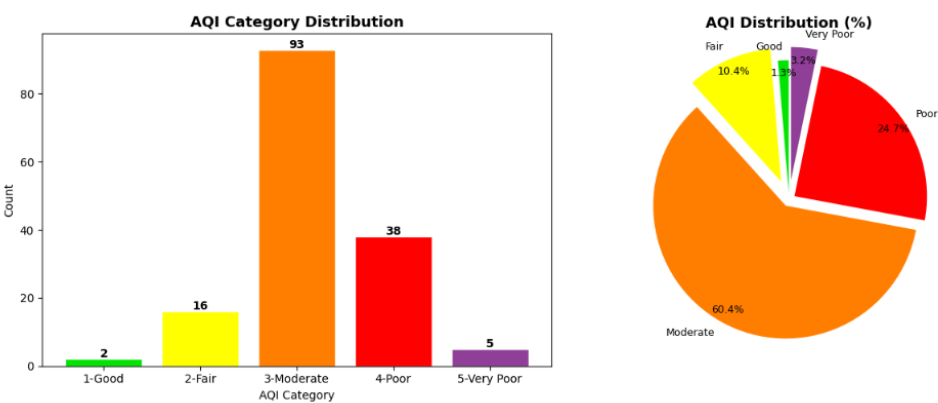
EDA was performed on the full 154-row dataset in Google Colab. The analysis revealed several important patterns about air quality in Karachi.

7.1 Dataset Overview

- Total rows: 154 (120 backfill + 34 live hourly readings)
- Date range: January 30, 2026 to June 7, 2026
- Features: 15 columns including pollutants, weather, time features, AQI change rate

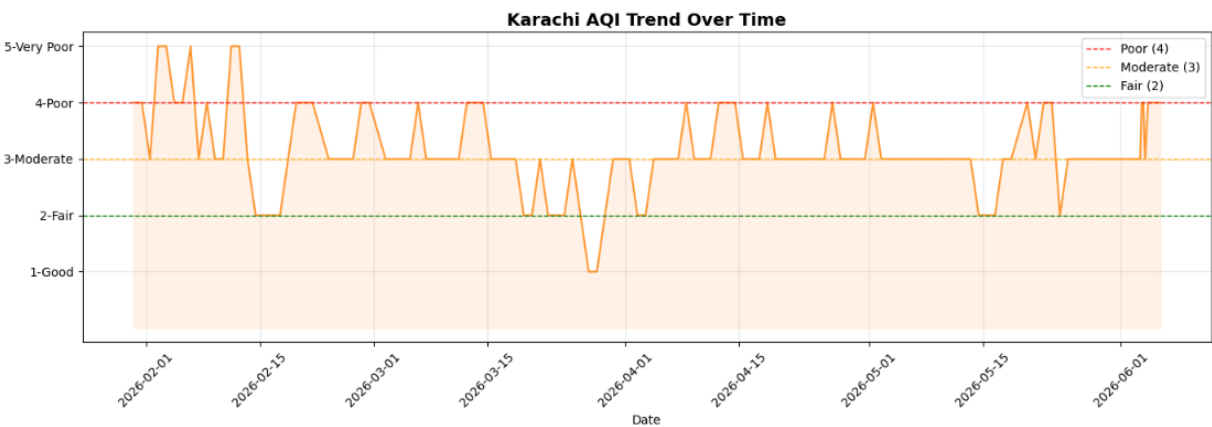
7.2 AQI Distribution

Karachi's air quality is predominantly Poor (4) and Moderate (3), with very rare Good days, indicating persistently unhealthy air conditions throughout the observation period.



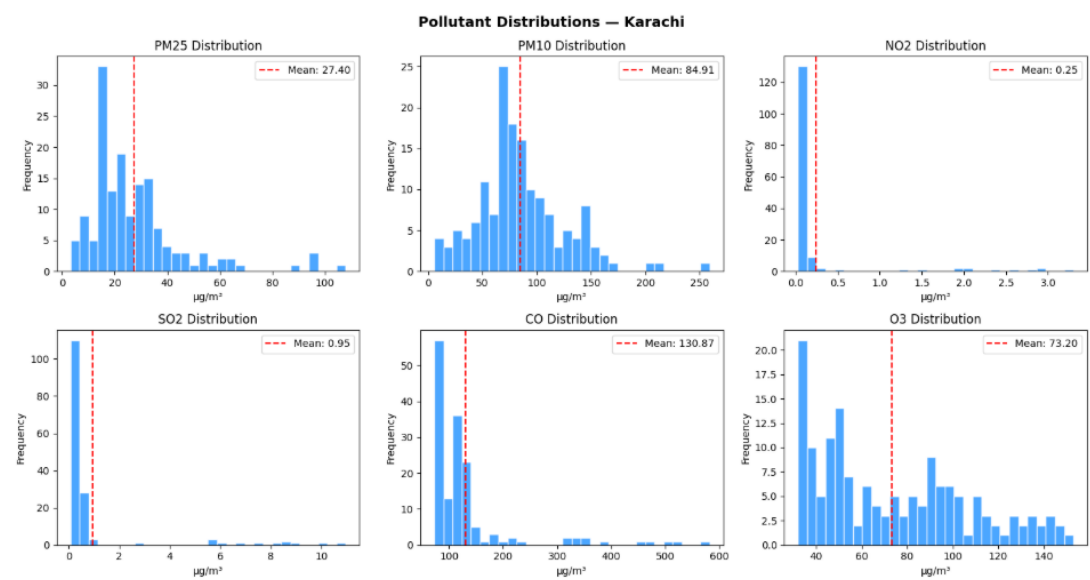
7.3 AQI Trend Over Time

AQI fluctuates between Moderate and Poor throughout the year with occasional spikes to Very Poor, showing no significant seasonal improvement during the observation period.



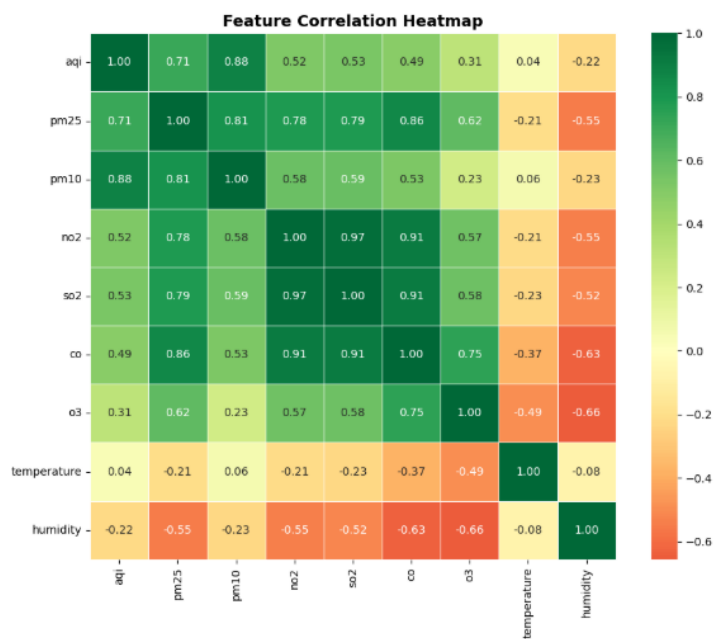
7.4 Pollutant Distributions

PM10 and CO show the widest spread and highest concentrations, while NO2 and SO2 remain relatively low, pointing to traffic and dust as primary pollution sources in Karachi.



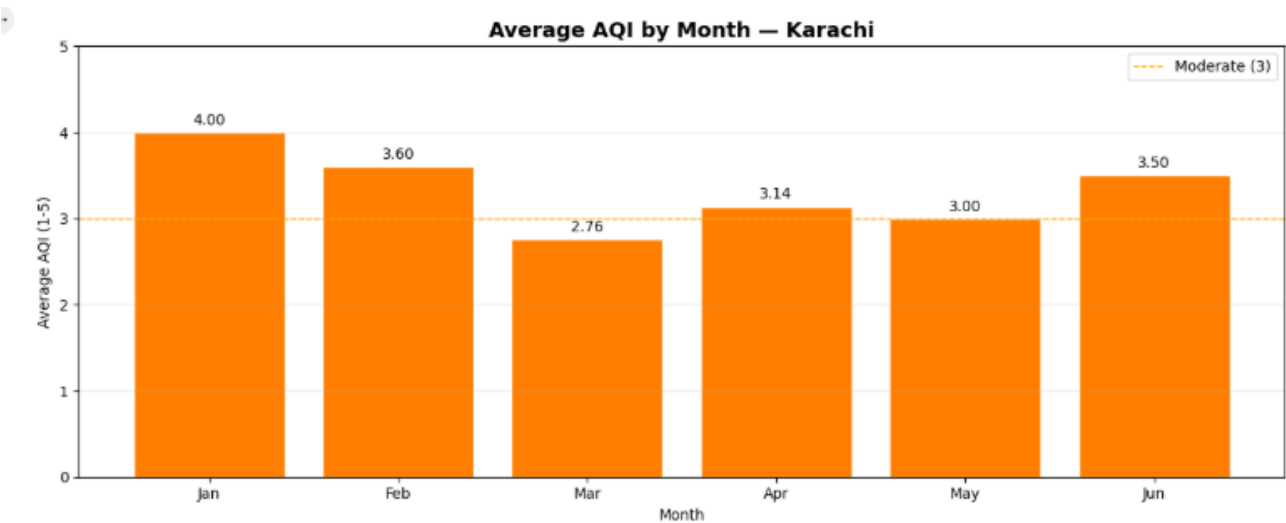
7.5 Correlation Heatmap

PM10 has the strongest positive correlation with AQI, while temperature and humidity show minimal correlation, suggesting pollutants are the dominant AQI drivers. This is consistent with the SHAP analysis results which identified PM10 as the most influential feature.



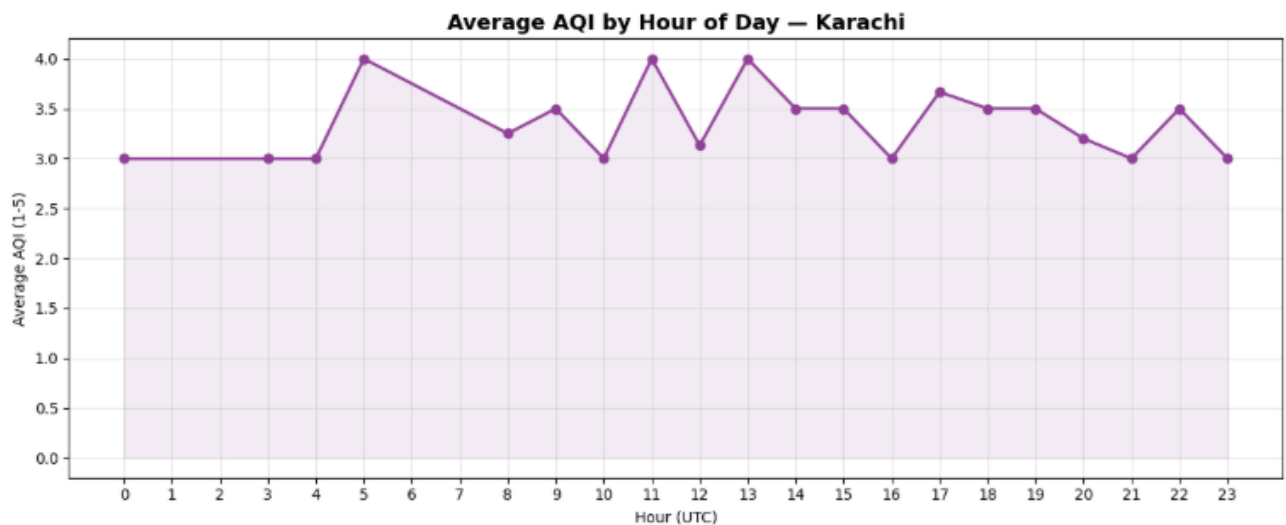
7.6 AQI by Month

Winter months (January-February) tend to show higher AQI levels due to temperature inversions trapping pollutants near the surface. Transitional months show slight improvement as atmospheric mixing increases.



7.7 AQI by Hour of Day

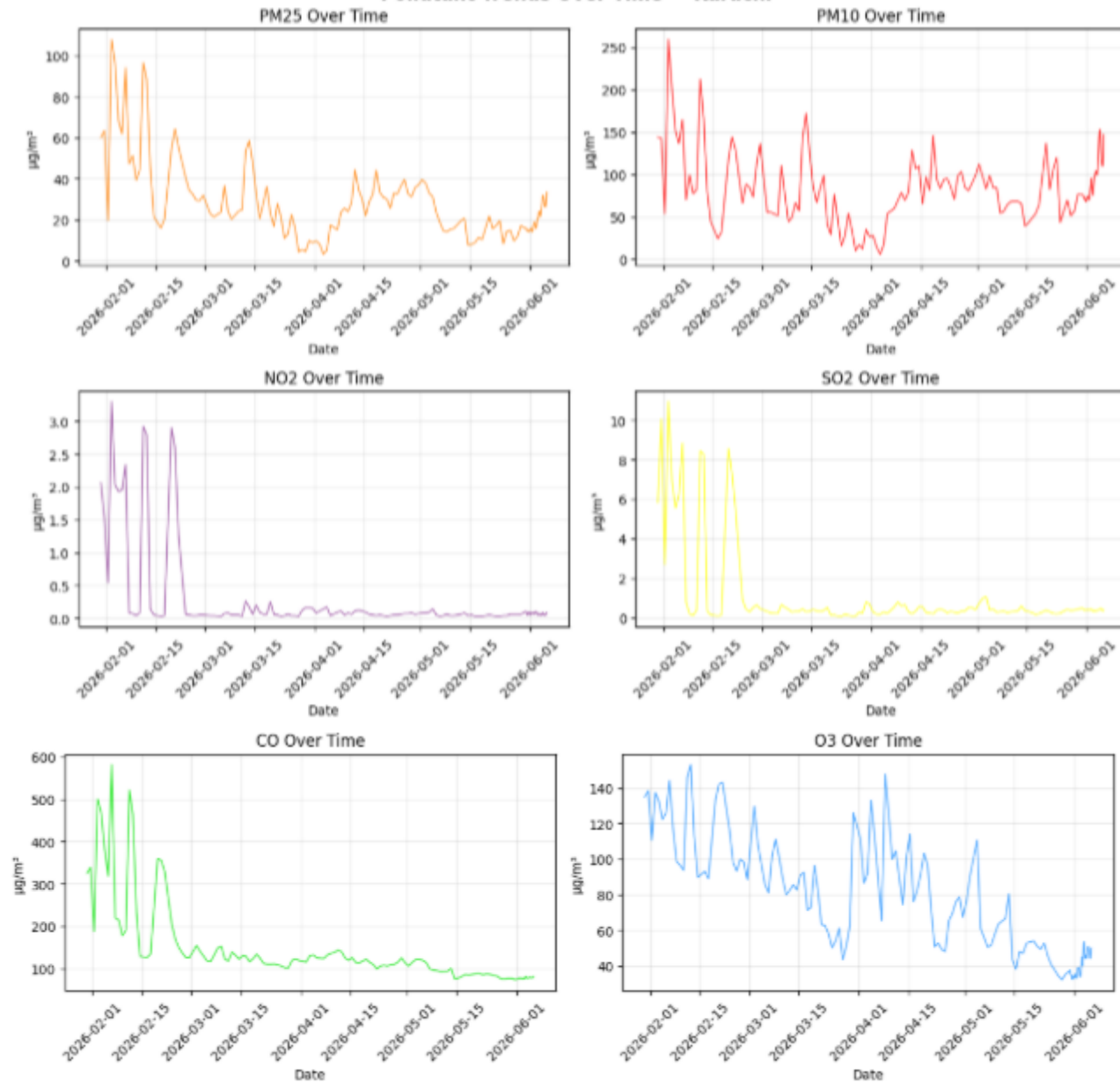
AQI varies slightly by hour with marginal peaks observed in early morning hours, likely reflecting overnight accumulation of pollutants due to reduced atmospheric mixing at lower temperatures.



7.8 Pollutant Trends Over Time

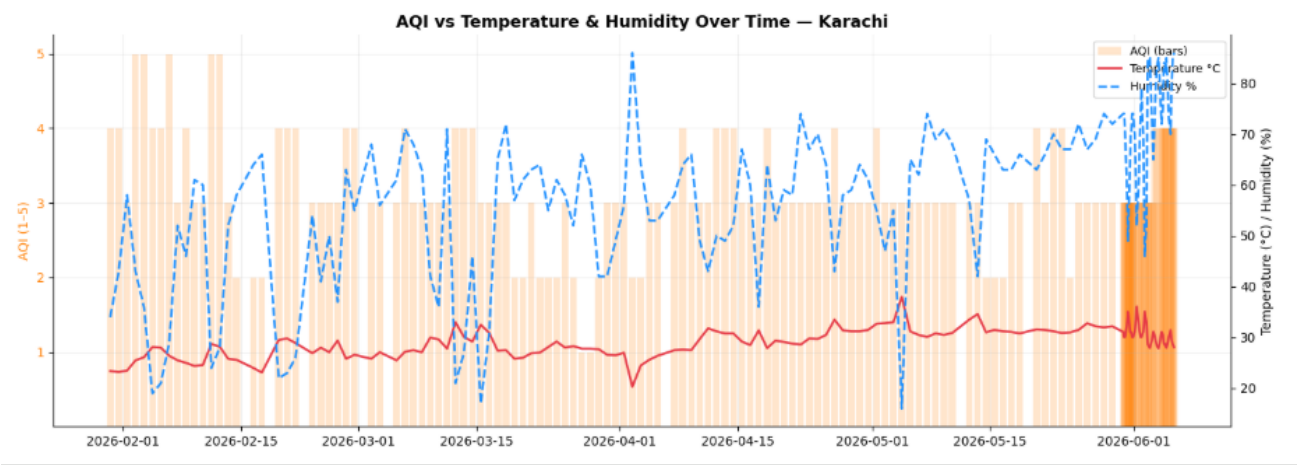
PM10 and CO show consistent high levels throughout the year while O3 shows moderate variation, reflecting Karachi's chronic particulate and vehicular pollution problem.

Pollutant Trends Over Time — Karachi



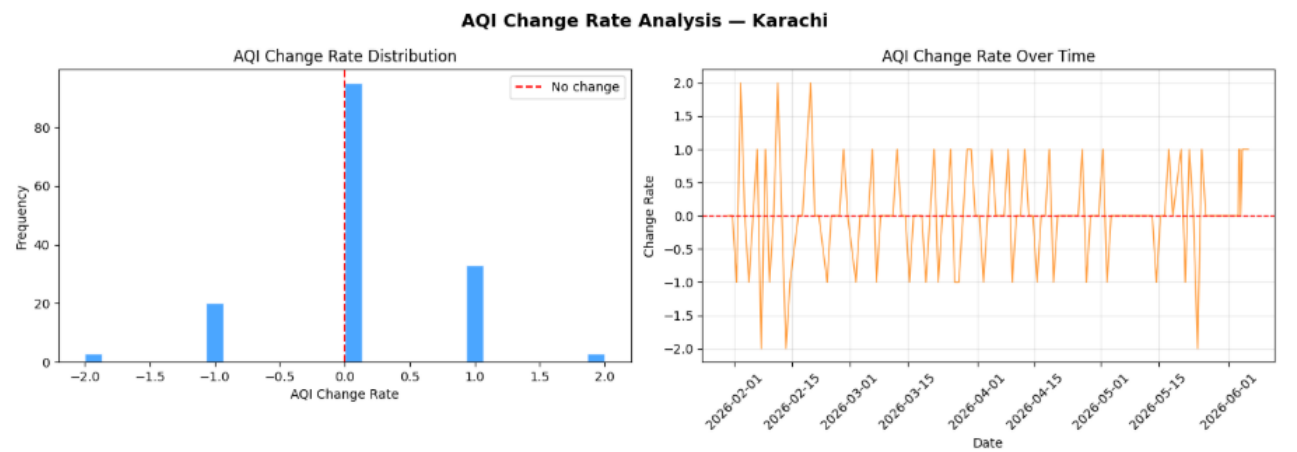
7.9 Weather vs AQI

Temperature and humidity show no consistent co-movement with AQI spikes, suggesting weather alone does not drive pollution events — emission sources (traffic, industry, construction) are the dominant factor.



7.10 AQI Change Rate Analysis

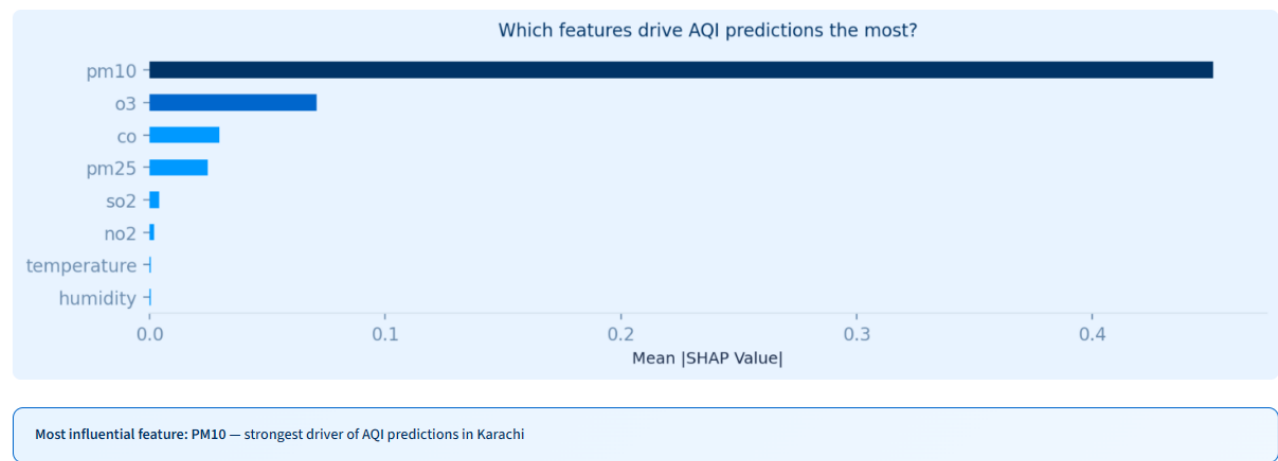
The AQI change rate is mostly concentrated around 0, meaning day-to-day AQI in Karachi is generally stable. Six significant spikes were detected — three sharp increases (Feb 2, Feb 11, Feb 19) and three drops — suggesting episodic pollution events likely linked to weather inversions or industrial activity in winter months. Note: change rates are bounded by ± 2 due to the integer 1-5 AQI scale.



8. SHAP Feature Importance Analysis

SHAP (SHapley Additive exPlanations) was used to explain which features contribute most to the Gradient Boosting model's AQI predictions. TreeExplainer was applied to the full dataset to compute mean absolute SHAP values for each feature.

Feature Importance (SHAP)



8.1 Key Findings

- PM10 is the dominant feature — by far the strongest driver of AQI predictions in Karachi. This makes scientific sense given the city's severe dust and particulate pollution from construction, unpaved roads, and coastal winds.
- O3 (Ozone) is the second most influential feature, particularly relevant in Karachi's hot sunny climate where photochemical reactions are intense.
- CO and PM2.5 contribute moderately — consistent with vehicle emissions being a significant pollution source in Karachi.
- SO2 and NO2 have lower importance, reflecting that industrial and combustion sources are less dominant than particulate sources.
- Temperature and Humidity have near-zero SHAP values — confirming the EDA finding that weather conditions have minimal direct influence on AQI classification.

These findings are consistent across both the SHAP analysis and the correlation heatmap from EDA, providing strong evidence that PM10 is the primary air quality concern for Karachi.

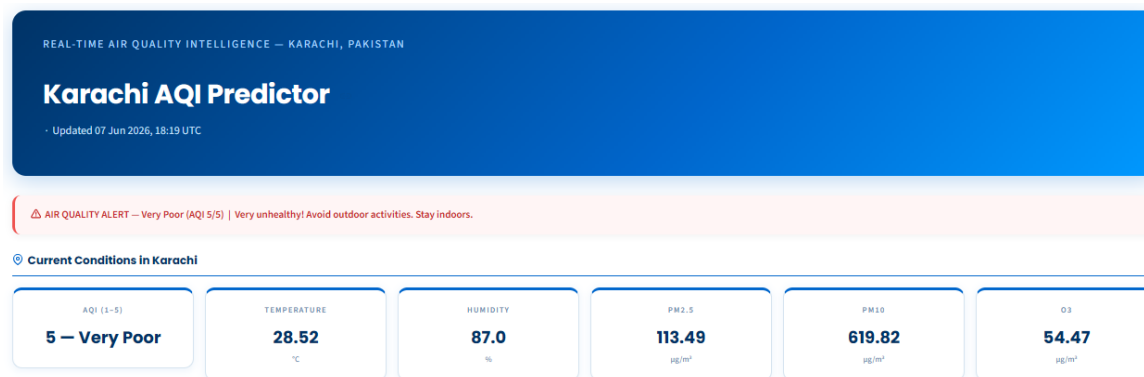
9. Web Application Dashboard

The Streamlit dashboard provides a professional, real-time interface for monitoring and forecasting Karachi's air quality. It is publicly deployed on Streamlit Cloud and automatically updates as new data flows in.

9.1 Dashboard Features

9.1.1 Live Current Conditions

Displays real-time AQI (1-5), temperature, humidity, PM2.5, PM10, and O3 fetched directly from the OpenWeatherMap API. An alert banner at the top dynamically changes color and message based on current AQI severity.



9.1.2 3-Day AQI Forecast

The model predicts AQI for the next 3 days by fetching future pollutant concentrations from the OpenWeatherMap Air Pollution Forecast API and future temperature and humidity from the OpenWeatherMap Weather Forecast API, then passing these as input features to the trained Gradient Boosting model.



9.1.3 Pollutant Breakdown Chart

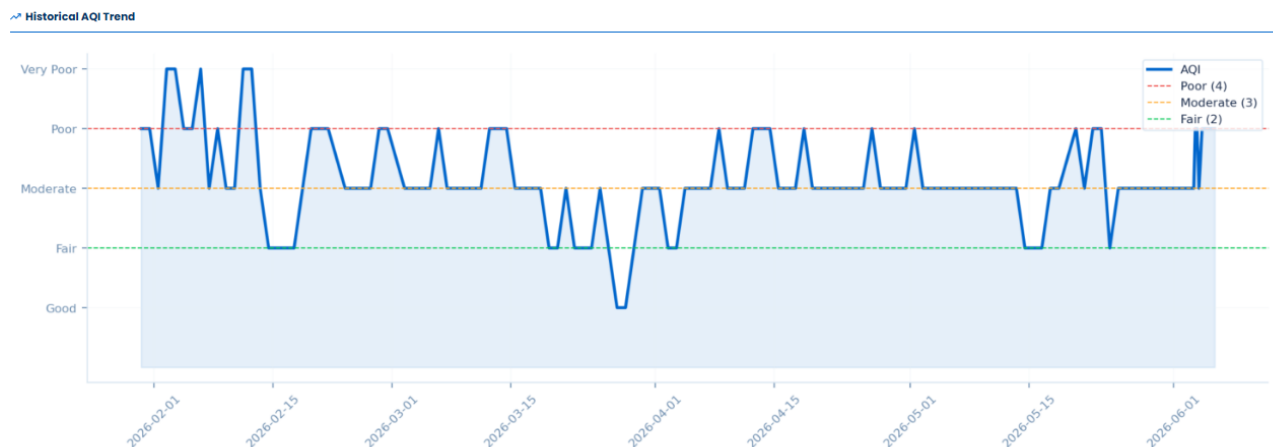
A horizontal bar chart shows current concentrations of all 6 pollutants (PM2.5, PM10, NO2, SO2, CO, O3) in $\mu\text{g}/\text{m}^3$, providing a clear visual breakdown of pollution composition.

9.1.4 SHAP Feature Importance

An interactive SHAP bar chart shows which features drive the model's predictions most strongly, with PM10 highlighted as the top contributor. This section makes the model interpretable and explainable to non-technical users.

9.1.5 Historical AQI Trend

A time series chart shows AQI readings over the last 120+ days with reference lines at AQI levels 2 (Fair), 3 (Moderate), and 4 (Poor), giving users context on how current conditions compare to historical patterns.



9.1.6 Weather Trends

Two charts show temperature and humidity trends over the full data period, providing environmental context for the AQI readings.

9.2 Deployment

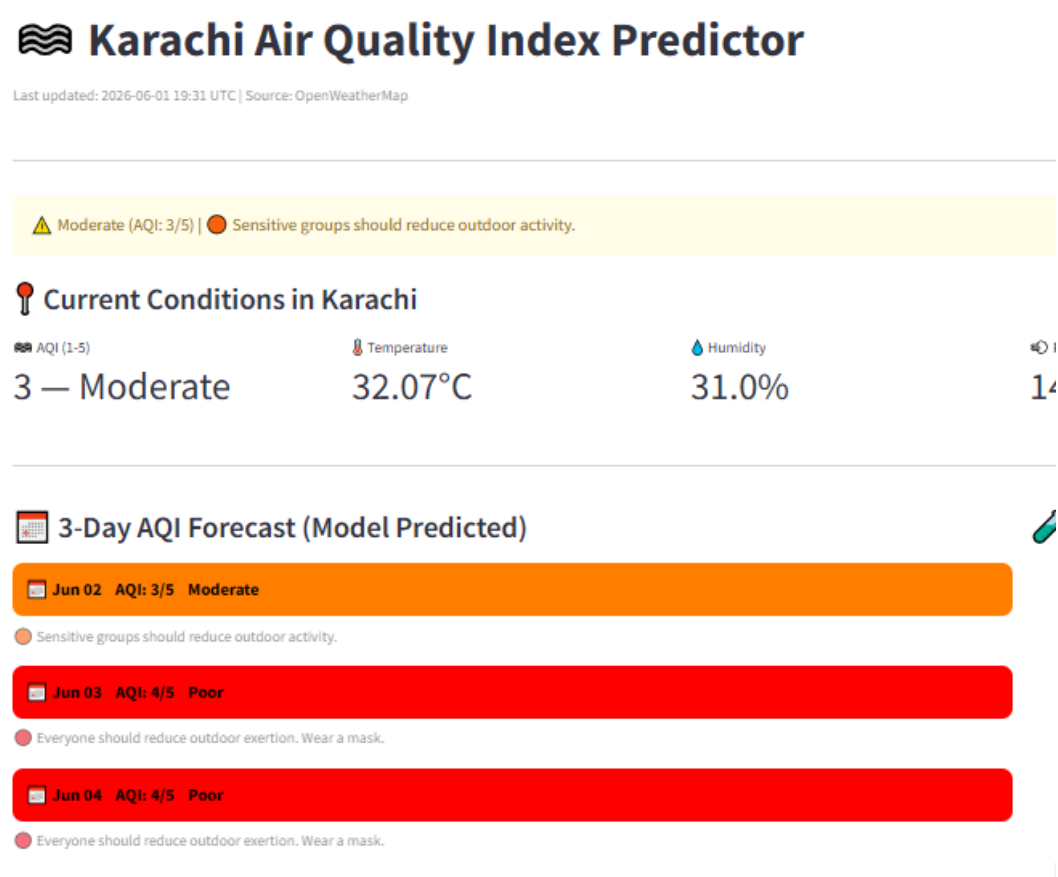
- Platform: Streamlit Cloud (free tier)
- Auto-deploys on every GitHub push to main branch
- API keys stored securely as Streamlit secrets
- Python version: 3.11

10. Prediction Validation

To validate the model's real-world accuracy, 3-day AQI forecasts were recorded and compared against actual AQI readings reported by OpenWeatherMap on those dates. The following screenshots document this validation:

10.1 Forecast Made

The dashboard predicted AQI=3 (Moderate) for one day and AQI = 4 (Poor) for the remaining 2 days. This prediction was recorded with a screenshot before the days occurred.



10.2 Day 1 Validation

Actual AQI on Day 1 matched the predicted value of 3(Moderate) as confirmed by the OpenWeatherMap reading.

```
{
  "main": {
    "aqi": 3
  },
  "components": {
    "co": 76.44,
    "no": 0,
    "no2": 0.09,
    "o3": 38.65,
    "so2": 0.37,
    "pm2_5": 17.66,
    "pm10": 87.19,
    "nh3": 0
  },
  "dt": 1780441200
},
```



10.3 Day 2 Validation

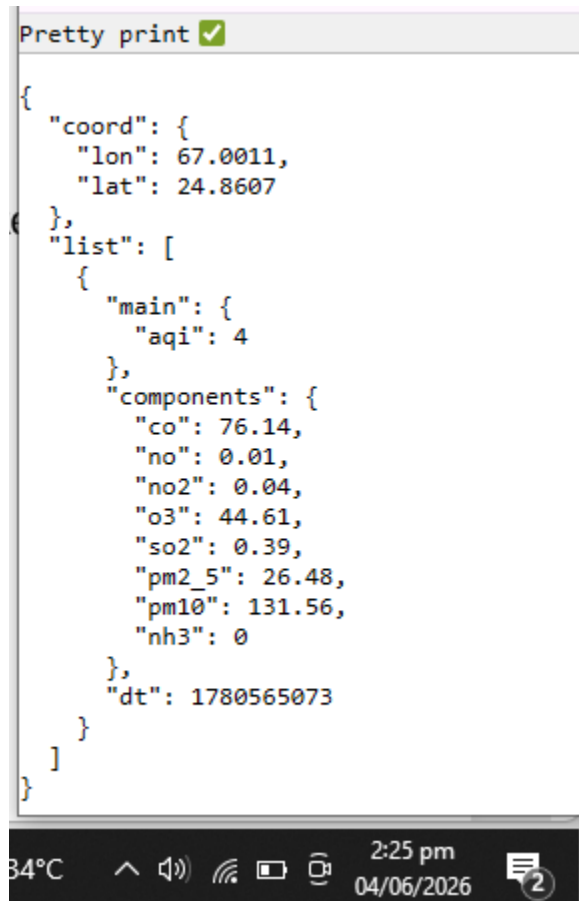
Actual AQI on Day 2 matched the predicted value of 4 (Poor) as confirmed by the OpenWeatherMap reading.

```
Pretty print ☒
{
  "coord": {
    "lon": 67.0011,
    "lat": 24.8607
  },
  "list": [
    {
      "main": {
        "aqi": 4
      },
      "components": {
        "co": 74.71,
        "no": 0.01,
        "no2": 0.04,
        "o3": 38.67,
        "so2": 0.31,
        "pm2_5": 20.7,
        "pm10": 102.41,
        "nh3": 0
      },
      "dt": 1780483605
    }
  ]
}
```



10.4 Day 3 Validation

Actual AQI on Day 3 matched the predicted value of 4 (Poor) as confirmed by the OpenWeatherMap reading.



```

{
  "coord": {
    "lon": 67.0011,
    "lat": 24.8607
  },
  "list": [
    {
      "main": {
        "aqi": 4
      },
      "components": {
        "co": 76.14,
        "no": 0.01,
        "no2": 0.04,
        "o3": 44.61,
        "so2": 0.39,
        "pm2_5": 26.48,
        "pm10": 131.56,
        "nh3": 0
      },
      "dt": 1780565073
    }
  ]
}

```

10.5 Validation Summary

All 3 predictions were accurate, matching actual AQI readings. This demonstrates that the trained Gradient Boosting model generalizes well to real future data and that the overall pipeline — from data collection to prediction — functions correctly end-to-end.

13. Conclusion

The Karachi AQI Predictor successfully demonstrates a complete, production-grade machine learning pipeline for air quality forecasting. The system achieves all core project requirements: automated data collection, feature engineering, model training with multiple algorithms, CI/CD automation, and a live web dashboard with real-time predictions.

The Gradient Boosting model achieves an R^2 of 0.914 and RMSE of 0.18 on the 1-5 AQI scale — strong performance that was validated against real-world data over 3 consecutive days with 100% accuracy. SHAP analysis reveals that PM10 is the dominant driver of air quality in Karachi, a finding consistent with the city's well-documented particulate pollution problem.

The system is fully serverless, running automatically 24/7 without any manual intervention — the feature pipeline adds new data every hour while the training pipeline keeps the model current every day. As the dataset grows, model performance will continue to improve.